

SHEBOYGAN COUNTY AP, WI, USA

WMO: **726425**

Lat: **43.769N** Lon: **87.851W** Elev: **746** StdP: **14.30** Time Zone: **-6.00 (NAC)** Period: **99-19** WBAN: **04841**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-5.1	-0.3	-14.4	2.6	-4.2	-9.8	3.3	0.3	26.2	20.5	23.9	21.0	10.2	280	0.581	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	20.1	88.1	73.9	84.3	71.5	81.8	70.0	75.7	84.4	73.9	81.8	72.1	79.3	11.6	220	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	72.9	125.4	81.3	71.4	119.2	79.1	69.7	112.4	77.1	39.7	84.4	37.9	81.8	36.3	79.3	84.4	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	24.1	20.6	18.5	DB	-11.2	92.6	5.2	3.6	-15.0	95.2	-18.0	97.4	-21.0	99.4	-24.8	102.1	(4)
(5)				WB	-11.7	78.9	5.0	2.0	-15.4	80.3	-18.3	81.5	-21.1	82.6	-24.8	84.1	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	45.8	20.1	21.7	32.5	43.6	54.4	64.5	70.0	68.0	61.2	49.3	37.5	25.5	(6)
(7)			DBStd	19.53	11.45	10.91	10.55	8.07	7.99	7.02	6.07	5.48	7.51	8.27	9.13	10.73	(7)
(8)			HDD50	3807	927	792	553	225	43	1	0	0	5	117	385	759	(8)
(9)			HDD65	7439	1392	1211	1009	643	341	93	21	30	159	492	824	1224	(9)
(10)			CDD50	2286	0	0	9	33	180	437	621	559	342	95	10	0	(10)
(11)			CDD65	443	0	0	0	0	13	79	177	124	46	4	0	0	(11)
(12)			CDH74	3802	0	0	2	21	152	722	1571	930	364	40	0	0	(12)
(13)			CDH80	1035	0	0	0	2	30	193	498	225	80	7	0	0	(13)
(14)	Wind		WSAvg	8.6	10.1	10.0	9.4	10.2	8.8	7.4	6.8	6.3	6.9	8.4	9.2	9.7	(14)
(15)	Precipitation		PrecAvg	32.2	1.5	1.3	1.9	3.2	3.8	4.0	3.5	3.3	2.9	2.7	2.1	1.9	(15)
(16)			PrecMax	39.5	3.3	3.9	4.2	5.0	10.0	10.3	7.9	6.1	5.2	5.4	5.6	4.0	(16)
(17)			PrecMin	24.6	0.4	0.1	0.3	1.1	0.9	1.3	1.3	1.0	0.6	0.7	0.4	0.7	(17)
(18)			PrecStd	3.7	0.9	0.8	1.1	1.2	2.0	2.1	1.5	1.4	1.3	1.3	1.3	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		47.6	51.7	68.3	76.9	83.8	90.3	93.4	90.6	87.1	79.3	66.3	53.6	(19)
(20)			MCWB		42.3	45.5	59.1	62.2	69.6	74.1	77.2	74.9	72.2	66.0	56.3	50.6	(20)
(21)		2%	DB		41.8	44.3	57.9	69.3	78.6	84.9	89.0	85.0	81.8	72.3	60.1	46.2	(21)
(22)			MCWB		38.0	40.0	52.1	56.8	65.7	70.8	75.0	72.7	69.8	62.8	52.3	42.3	(22)
(23)		5%	DB		38.1	39.0	52.2	62.8	73.5	81.6	85.0	82.0	78.5	67.1	54.8	42.9	(23)
(24)			MCWB		35.8	35.8	46.4	53.0	62.7	69.5	72.3	70.5	67.9	59.5	49.3	40.3	(24)
(25)		10%	DB		35.8	36.1	46.3	57.2	69.8	78.5	82.1	79.2	74.5	63.1	51.0	39.1	(25)
(26)			MCWB		33.9	33.6	41.7	48.3	60.0	66.9	70.6	68.7	65.8	57.2	46.7	36.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		43.2	46.1	60.3	63.8	71.1	76.5	79.3	77.8	74.7	70.0	57.8	51.3	(27)
(28)			MCDB		46.6	50.5	67.5	72.8	81.7	87.2	89.6	87.4	83.1	76.9	64.7	52.2	(28)
(29)		2%	WB		38.5	40.3	53.2	58.8	67.5	73.3	76.5	74.9	72.0	64.8	53.7	43.7	(29)
(30)			MCDB		40.6	43.6	57.8	67.9	75.4	82.4	86.0	82.1	79.0	68.9	57.9	44.9	(30)
(31)		5%	WB		36.2	36.4	46.4	54.0	64.8	71.0	74.5	72.7	69.9	61.2	50.3	40.3	(31)
(32)			MCDB		38.2	39.0	50.8	60.9	72.0	79.2	82.6	79.3	75.8	66.0	54.5	42.6	(32)
(33)		10%	WB		33.7	33.6	42.0	49.7	61.5	68.6	72.7	70.7	67.7	57.8	46.7	37.0	(33)
(34)			MCDB		35.7	35.5	46.4	56.5	68.2	75.8	80.1	76.8	73.1	62.3	50.0	39.1	(34)
(35)	Mean Daily Temperature Range		MDBR		13.6	14.9	15.3	17.5	19.3	19.4	20.1	19.3	20.4	17.2	15.0	12.9	(35)
(36)		5% DB	MCDBR		14.3	16.3	22.1	26.7	25.2	23.2	22.6	22.2	24.0	24.1	21.5	15.0	(36)
(37)			MCWBR		12.2	13.0	16.0	16.9	15.1	12.8	11.5	11.5	13.3	15.6	15.7	12.8	(37)
(38)		5% WB	MCDBR		13.8	15.9	21.3	25.1	23.4	21.0	20.3	19.6	20.9	20.5	19.8	14.1	(38)
(39)			MCWBR		12.2	12.9	16.2	17.3	14.9	12.7	11.4	11.0	12.7	15.3	15.5	12.8	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.281	0.314	0.337	0.383	0.405	0.423	0.415	0.408	0.382	0.340	0.304	0.282	(40)
(41)		taud			2.430	2.322	2.378	2.292	2.281	2.278	2.307	2.316	2.377	2.486	2.520	2.475	(41)
(42)		Ebn at Noon			272	278	286	279	275	270	271	268	267	266	261	260	(42)
(43)		Edh at Noon			23	31	34	40	41	42	40	38	33	26	22	20	(43)
(44)	All-Sky Solar Radiation	RadAvg			472	780	1152	1439	1709	1899	1947	1664	1332	843	526	371	(44)
(45)		RadStd			62	84	115	136	141	133	87	107	108	92	46	38	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page